



**Department of Electrical, Computer
& Biomedical Engineering**
Faculty of Engineering & Architectural Science

Course Title:	Low Power Digital Integrated Circuits
Course Number:	ELE734
Semester/Year (e.g. F2016)	F2025

Instructor:	Dr. Andy Gean Ye
Lab Report Number	Lab #4 - Lab
Lab Title	Lab 4 - Lab: CMOS 1-bit full adder

Due Date	December 04, 2025
Submission Date	December 02, 2025

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*By signing above you attest that you have contributed to this written lab report and confirm that all work you have contributed to this lab report is your own work. Any suspicion of copying or plagiarism in this work will result in an investigation of Academic Misconduct and may result in a "0" on the work, an "F" in the course, or possibly more severe penalties, as well as a Disciplinary Notice on your academic record under the Student Code of Academic Conduct, which can be found online at: <http://www.ryerson.ca/senate/current/pol60.pdf>

Lab 4 : Pre - Lab:

1)

A	B	C_{in}	C_{out} (Carry Out)	S (Sum)
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Table 1: Truth Table for a 1-bit full adder

From the following Truth Table for a 1-bit full adder, we can determine that the boolean logic to be:

$$S = A \oplus B \oplus C_{in}$$

$$C_{out} = A \cdot B + A \cdot C_{in} + B \cdot C_{in} = (A \cdot B) + (A \oplus B)C_{in} \text{ [Using the Factoring Rule]}$$

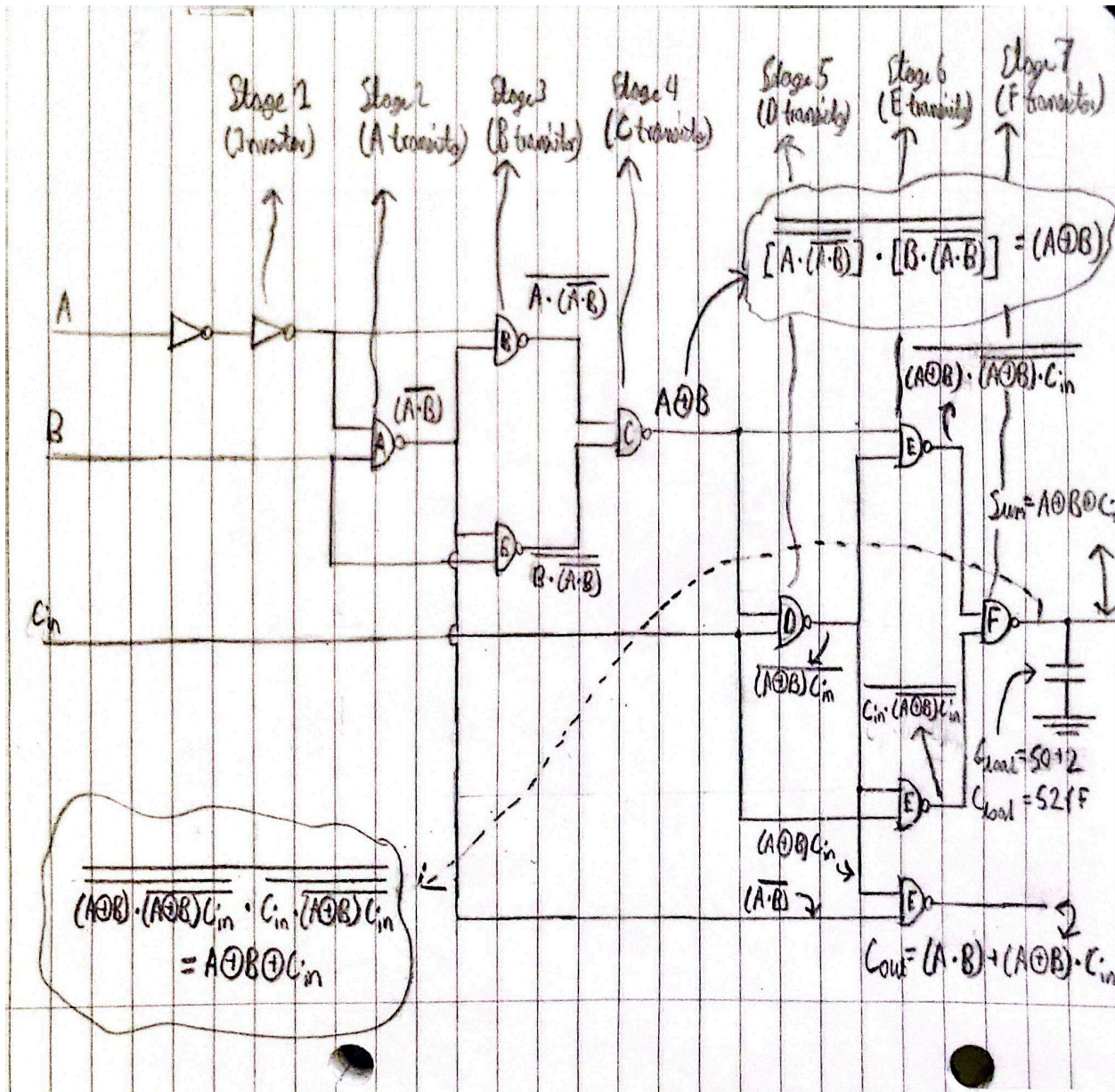


Figure 1: 1-bit Full Adder Gate Level Schematic

2)

Logic Effort Method:

Load Capacitance: $C_{load} = 50 \text{ fF} + 02 \text{ fF (last two digits of student number)} = 52 \text{ fF}$

Stage 1: $g_1 = 1$, $p_1 = 1$, $h_1 = \frac{2a}{1} [b_1 = 2]$

Stage 2: $g_2 = \frac{4}{3}$, $p_1 = 2$, $h_2 = \frac{3b}{a} [b_2 = 3]$

Stage 3: $g_3 = \frac{4}{3}$, $p_1 = 2$, $h_3 = \frac{c}{b} [b_3 = 1]$

$$\text{Stage 4: } g_4 = \frac{4}{3}, p_1 = 2, h_4 = \frac{2d}{c} [b_4 = 2]$$

$$\text{Stage 5: } g_5 = \frac{4}{3}, p_1 = 2, h_5 = \frac{3e}{d} [b_5 = 3]$$

$$\text{Stage 6: } g_6 = \frac{4}{3}, p_1 = 2, h_6 = \frac{f}{e} [b_6 = 1]$$

$$\text{Stage 7: } g_7 = \frac{4}{3}, p_1 = 2, h_7 = \frac{52}{f} [b_7 = 1]$$

Note that since the schematic consists of only the NAND gates, the unskewed NAND gate's logic effort is known to be 4/3(as found in Lab 3).

$$\sum_{i=1}^7 g_i = G = 1 * \frac{4}{3} * \frac{4}{3} * \frac{4}{3} * \frac{4}{3} * \frac{4}{3} * \frac{4}{3} = 5.619$$

$$\sum_{i=1}^7 b_i = B = 2 * 3 * 1 * 2 * 3 * 1 * 1 = 36$$

$$\sum h_i = H = \frac{a}{1} * \frac{b}{a} * \frac{c}{b} * \frac{d}{c} * \frac{e}{d} * \frac{f}{e} * \frac{52}{f} = 52$$

$$\text{Now, the formula for the Full Path Effort: } F = \prod_{i=1}^7 g_i h_i = \prod_{i=1}^7 g_i h_i b_i = G * H * B$$

$$F = 36 * 52 * 5.619 = \mathbf{10518.768}$$

$$\text{Now, the optimal stage effort(Target per-stage for electrical effort) is: } f = F^{1/N} = (10518.768)^{1/7} = \mathbf{3.755}$$

At each optimum stage, the path effort equation should satisfy: $g_i h_i b_i = f$

$$\text{Stage 1: } g_1 = 1, h_1 = \frac{2a}{1} \Rightarrow 1 * \frac{2a}{1} = 3.755 \rightarrow a = 1.877$$

$$\text{Stage 2: } g_2 = \frac{4}{3}, h_2 = \frac{3b}{a} \Rightarrow \frac{4}{3} * \frac{3b}{a} = 3.755 \rightarrow b = 1.762 [a = 1.877]$$

$$\text{Stage 3: } g_3 = \frac{4}{3}, h_3 = \frac{c}{b} \Rightarrow \frac{4}{3} * \frac{c}{b} = 3.755 \rightarrow c = 4.962 [b = 1.762]$$

$$\text{Stage 4: } g_4 = \frac{4}{3}, h_4 = \frac{2d}{c} \Rightarrow \frac{4}{3} * \frac{2d}{c} = 3.755 \rightarrow d = 6.987 [c = 4.962]$$

$$\text{Stage 5: } g_5 = \frac{4}{3}, h_5 = \frac{3e}{d} \Rightarrow \frac{4}{3} * \frac{3e}{d} = 3.755 \rightarrow e = 6.558 [d = 6.987]$$

$$\text{Stage 6: } g_6 = \frac{4}{3}, h_6 = \frac{f}{e} \Rightarrow \frac{4}{3} * \frac{f}{e} = 3.755 \rightarrow f = 18.467 [e = 6.558]$$

$$\text{Stage 7: } g_7 = \frac{4}{3}, h_7 = \frac{52}{f} \Rightarrow \frac{4}{3} * \frac{52}{f} = 3.754 \simeq 3.755 \rightarrow \text{CORRECT } [f = 18.467]$$

Width for the NAND gate:

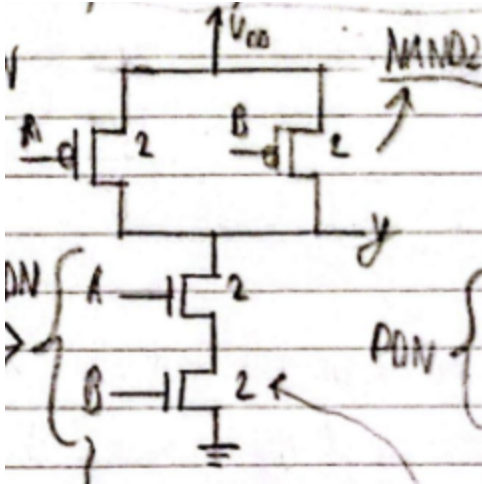


Figure 2: Unskewed NAND2 Gate Width view

Since, in a unskewed original NAND gate, the width size for the PMOS(Pull-up network) and NMOS(Pull-down network) are the same(width of '2'); the total capacitance held by the NAND transistor(input capacitance) is 4, where the distribution of the capacitance is equal as the PMOS holds $\frac{2}{4}$ and NMOS holds $\frac{2}{4}$. That means the width of the NMOS and PMOS are the same as the capacitance distribution is equal, both being half the NAND gate input capacitance.

Let, 1 Unit(Inverter Unit) $\rightarrow$ 3 C (PMOS+NMOS= 2+1 = 3):

NMOS and PMOS width for NAND 'a' is: $\frac{1.877}{2} = 0.939$	Width in Capacitance: $0.939 * 3 = 2.817 C$
NMOS and PMOS width for NAND 'b' is: $\frac{1.762}{2} = 0.881$	Width in Capacitance: $0.881 * 3 = 2.643 C$
NMOS and PMOS width for NAND 'c' is: $\frac{4.962}{2} = 2.481$	Width in Capacitance: $2.481 * 3 = 7.443 C$
NMOS and PMOS width for NAND 'd' is: $\frac{6.987}{2} = 3.494$	Width in Capacitance: $3.494 * 3 = 10.482 C$
NMOS and PMOS width for NAND 'e' is: $\frac{6.558}{2} = 3.279$	Width in Capacitance: $3.279 * 3 = 9.837 C$
NMOS and PMOS width for NAND 'f' is: $\frac{18.467}{2} = 9.234$	Width in Capacitance: $9.234 * 3 = 27.702 C$

Lab 4 : In - Lab:

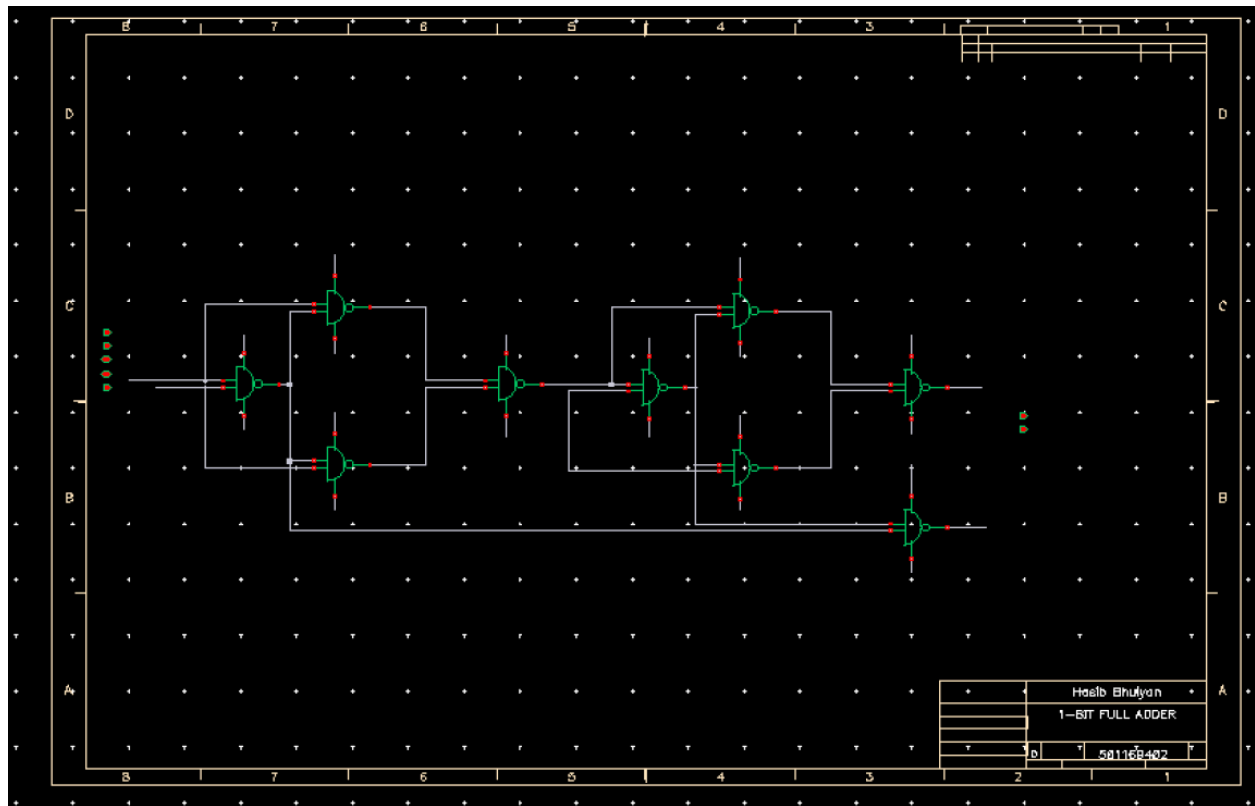


Figure 3: Lab 4: 1-bit FULL Adder Original Schematic Design

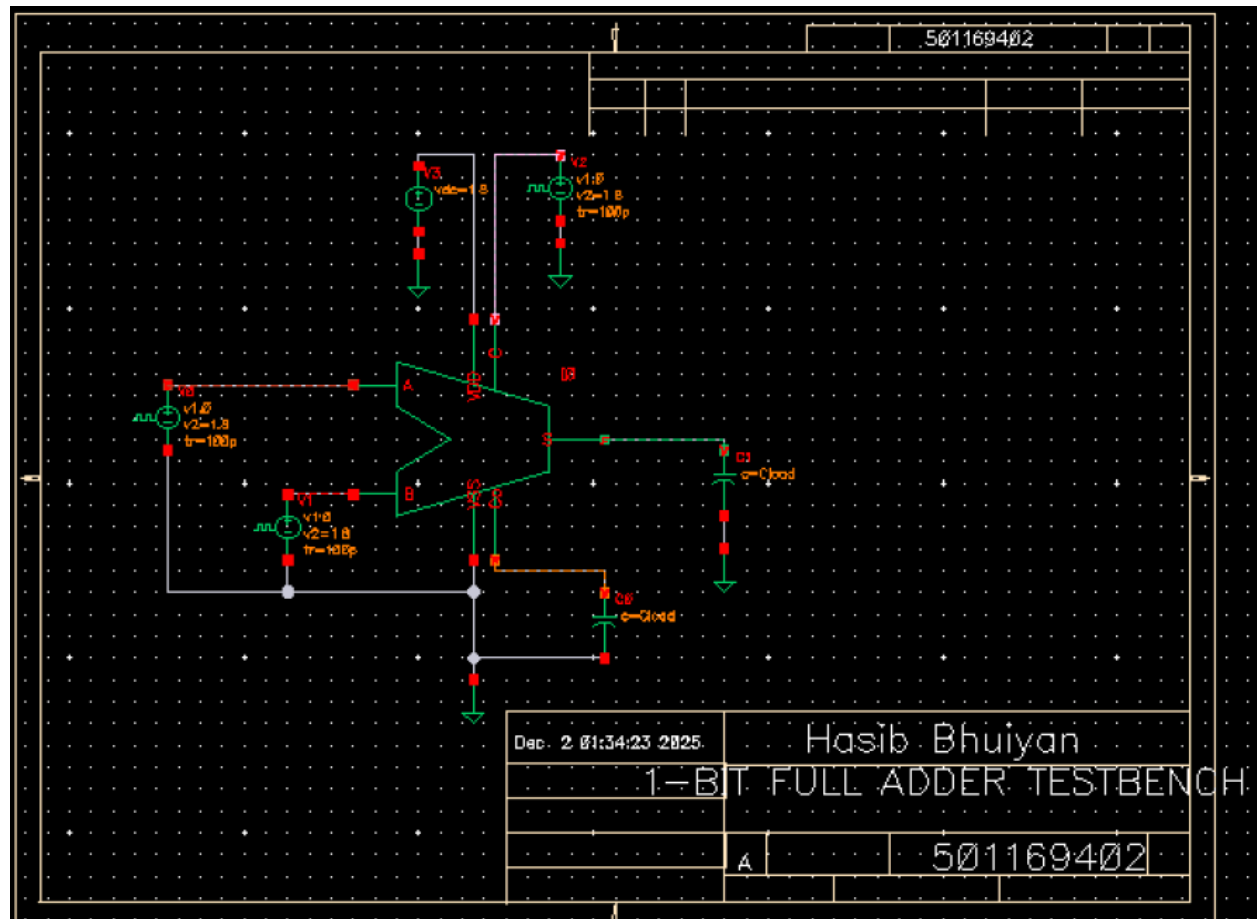


Figure 4: Lab 4: 1-bit FULL Adder TestBench Design

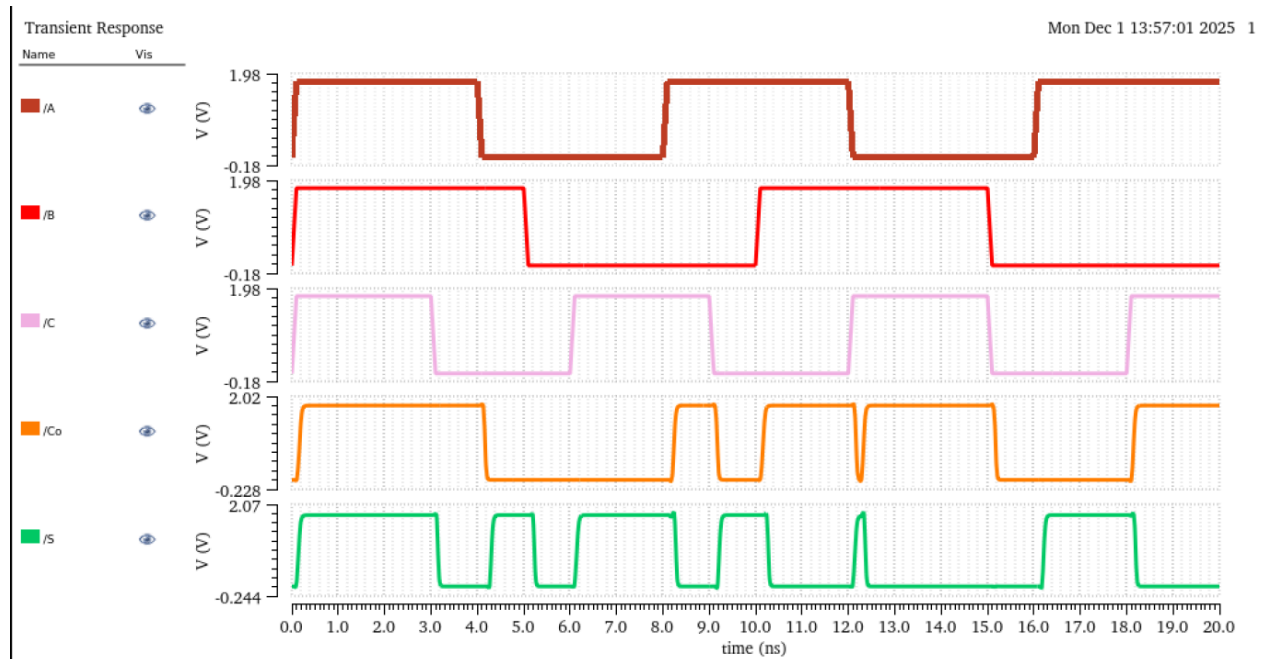


Figure 5: Lab 4: 1-bit FULL Adder TestBench Simulation (Pre-layout Transient Response)

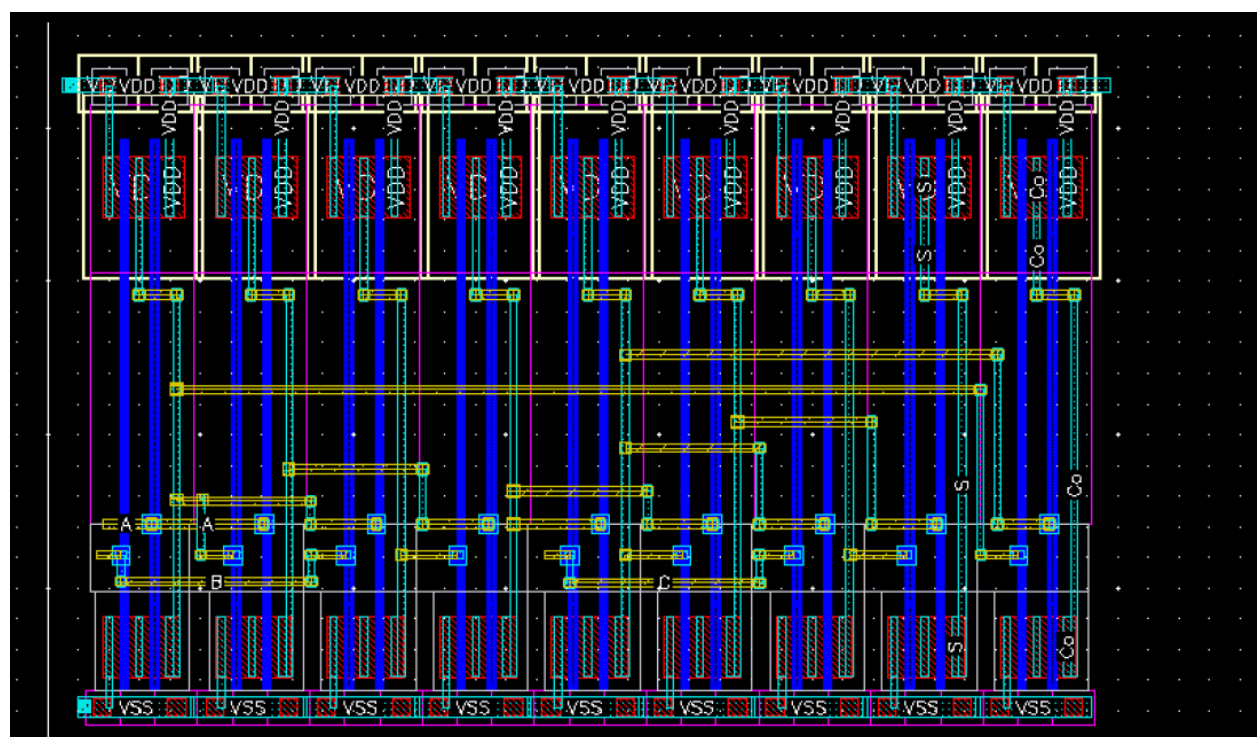


Figure 6: Lab 4: 1-bit FULL Adder LAYOUT Design

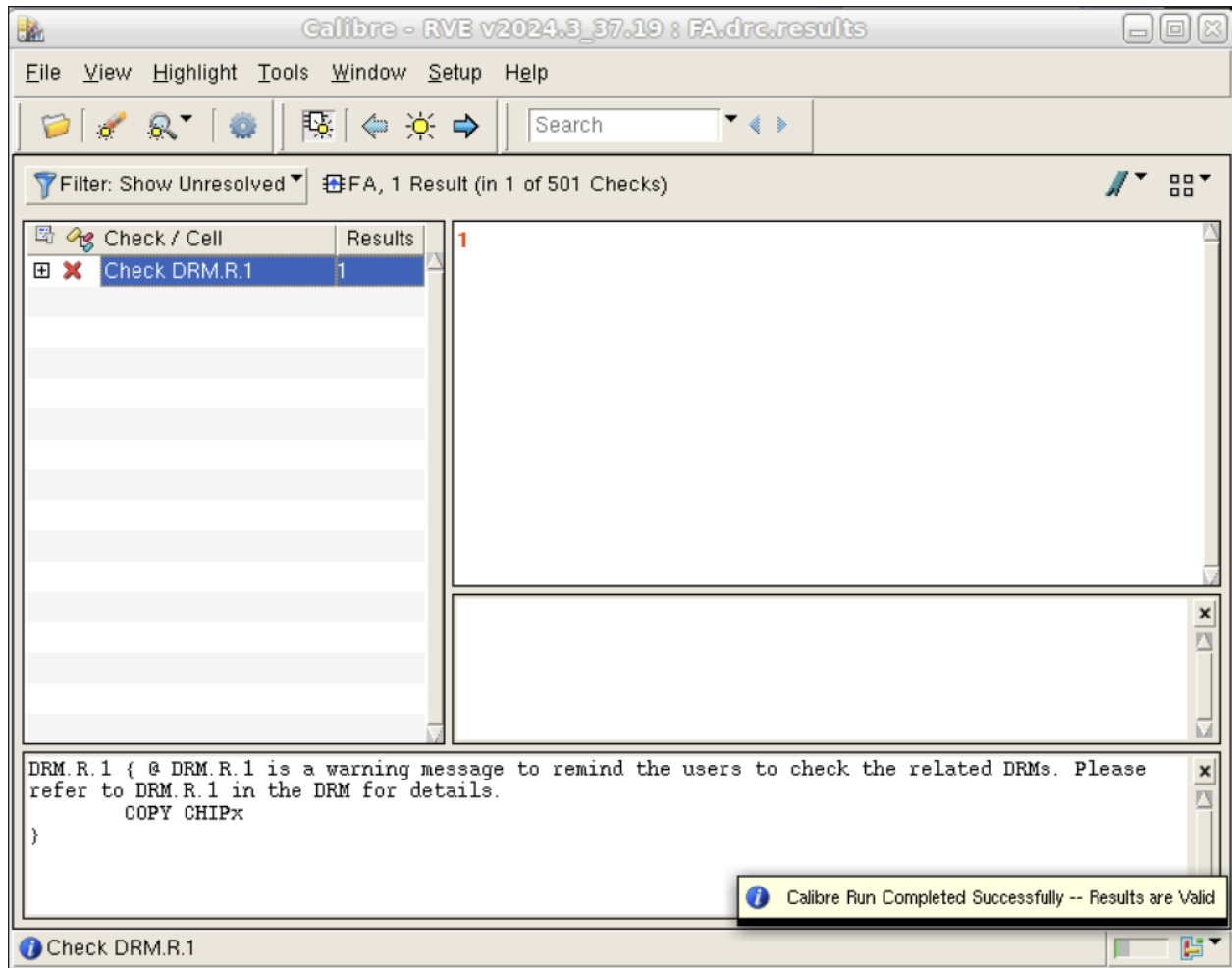


Figure 7: Lab 4: 1-bit FULL Adder LAYOUT nmDRC Check Results

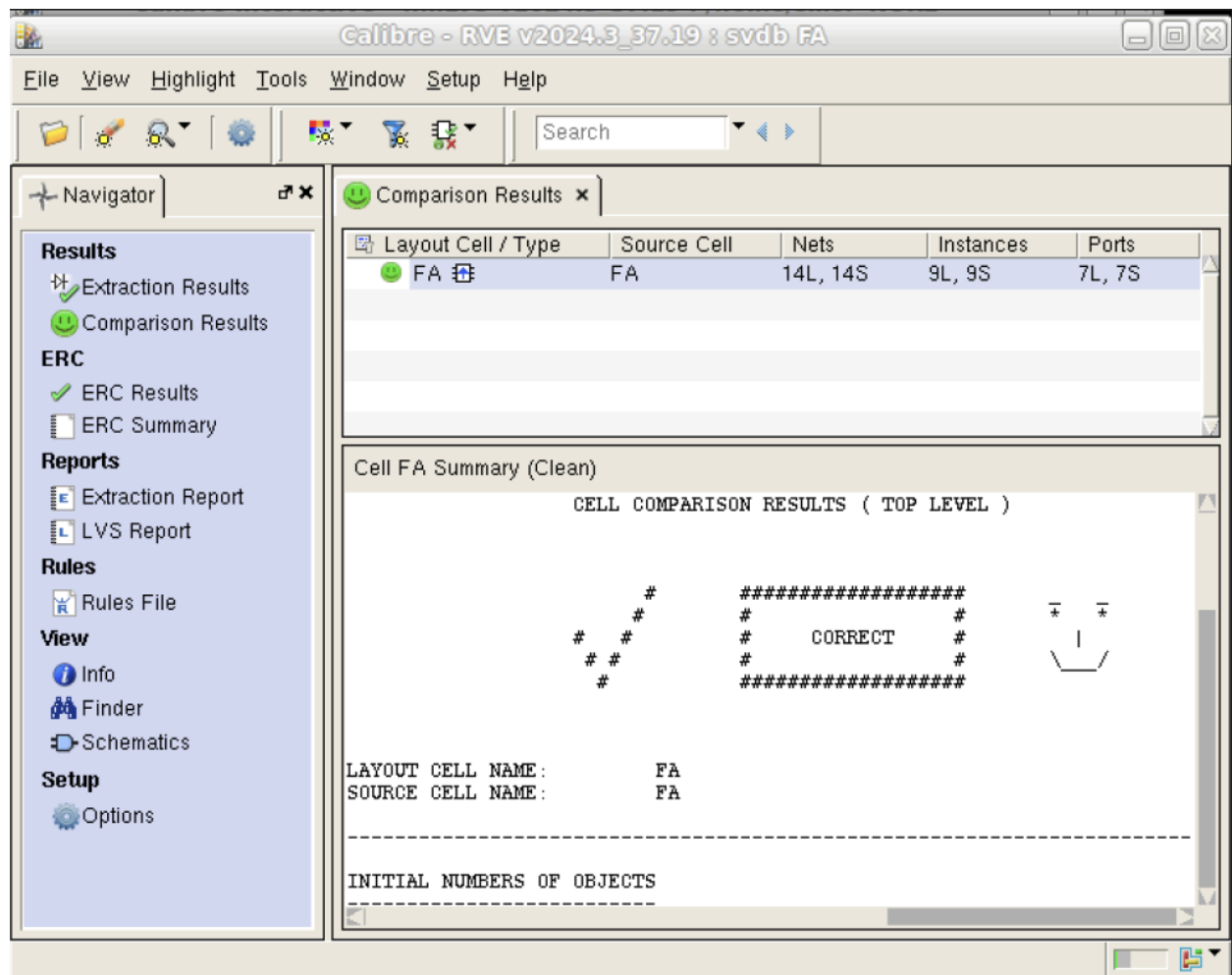


Figure 8: Lab 4: 1-bit FULL Adder LAYOUT nmLVS Check Results

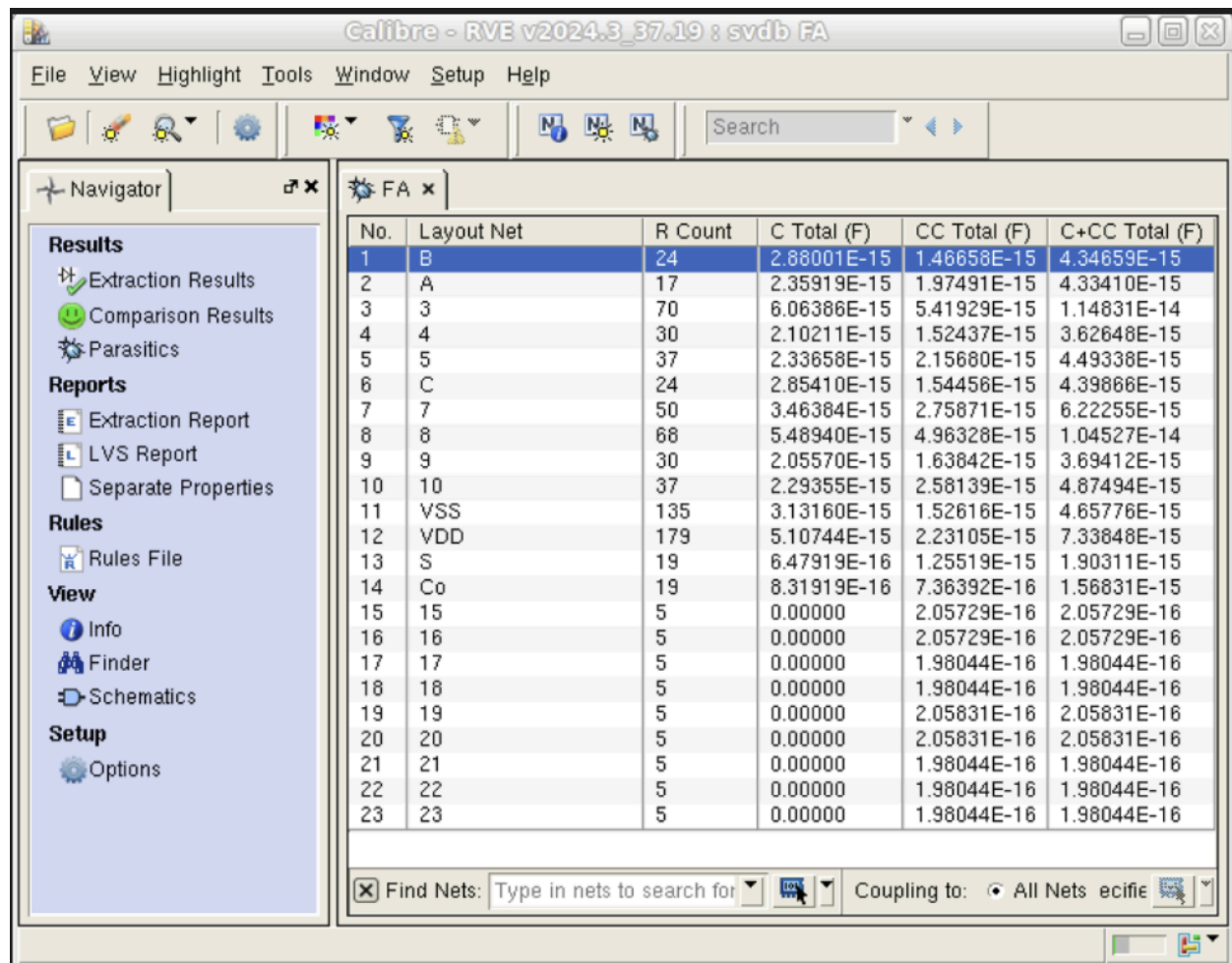


Figure 9: Lab 4: 1-bit FULL Adder LAYOUT PEX Extraction Numerical Results



Figure 10: Lab 4: 1-bit FULL Adder LAYOUT PEX Extraction VIEW Results



Figure 11: Lab 4: 1-bit FULL Adder LAYOUT PEX Extraction VIEW Results[ZOOMED]

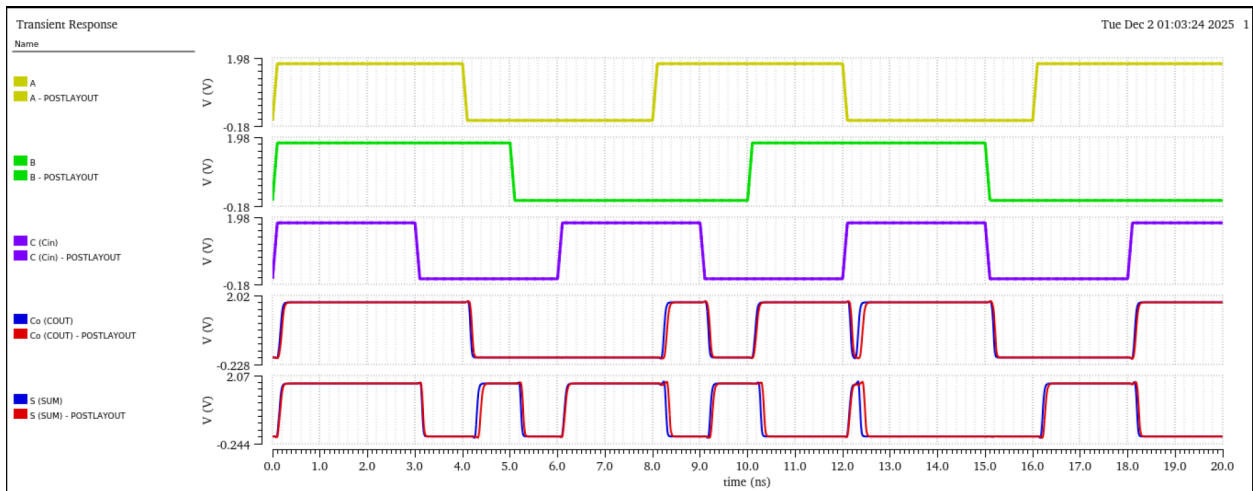


Figure 12: Lab 4: 1-bit FULL Adder TestBench Post-Layout Simulation (Pre-layout VS Post-layout Transient Response)

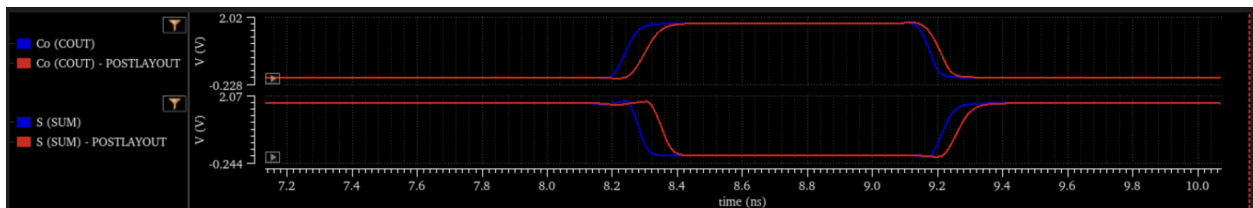


Figure 13: Lab 4: 1-bit FULL Adder TestBench Post-Layout Simulation[ZOOMED in to the actual Post-Layout Impact]

The lab #4 focuses on designing, sizing, and simulating a 1-bit full adder using CMOS logic based on the guidelines introduced within the manual. To begin with, the pre-lab relied entirely on theoretical analysis using logical effort to determine transistor widths for the gates within the SUM and COUT paths in the 1-bit full adder schematic. Do note that under these assumptions, parasitics, wiring effects, and any non-ideal cases were entirely ignored, resulting in ideal theoretical results with predetermined propagation delays and clean, symmetrical switching behaviour within NAND gates. These pre-lab predictions represent the theoretical expectations for what an ideally sized full-adder should achieve in this lab in terms of delay, noise margins, and behaviour.

The schematic simulations effectively apply realistic simulation calculations on the simulation testbench created. However it still excludes layout parasitics, and closely matches some of the behaviour within our pre-lab expectations (like the Truth Table for the 1-bit Full Adder and its boolean functions). The SUM output and COUT output both had the correct functional behaviour for all input combinations (as it could be seen in figure 5 and 12), with accurate boolean expression and values compared to the ideal theoretical values. The overall waveforms were sharp, well-defined, and consistent with ideal expectations and theory. At this stage, the minor differences from the pre-lab are minimal, some being the rise/fall time and different delays which the pre-lab totally ignores.

Now the most significant changes and impact appeared in the extracted post-layout simulations; this makes sense as external factors can play a role, for example the factors like parasitic resistances, interconnect capacitances, and real routing paths (layout routing factors) that were introduced. The post-layout SUM and COUT waveforms clearly show the expected behaviour; the shapes of the curves remain exactly the same as the pre-layout schematic results, except both signals are shifted, slightly stretched, and delayed in time for the post-layout curve. This verifies that the circuit still functions correctly, but reflects the very much expected slowdown caused by additional parasitic loading that shows up in a real physical layout. Thus, as said, the functional behaviour remains identical, and the timing deviations/delays are small (see figure 12 & 13), allowing us to understand that the general waveform structure is preserved.

As it could be seen, the consistency among the three stages; pre-lab, schematic, and extracted simulations, proves that the theoretical design, ideal simulations, and real layout work very much similarly and apply the same theory and behaviour. The minor timing deviations/delays observed in the post-layout results shows that the 1-bit full adder's real delays are negligible and also show that the full-adder design remains very much reliable when subjected to real-world parasitics.